

SECTION 6

MASS AND BALANCE

6.1 GENERAL

This section provides information required for helicopter loading and computing mass and balance.

It shall be the pilot's responsibility to make certain that:

- the helicopter is properly loaded so that the entire flight is conducted within the Center of Gravity Limits,
- all cargo is stowed and tied down properly so that in-flight shifting is impossible,
- proper tie-down equipment (i.e. ropes, belts, etc.) of sufficient strength has to be used. As many tie-down fittings as possible have to be used per single cargo item in order to reduce the individual load per fitting and to avoid inadvertent in-flight shifting.

6.1.1 Mass Definitions

Basic empty mass consists of the basic helicopter with required standard equipment, optional equipment, unusable fuel, and full operating fluids including transmission, gearbox and engine oils, hydraulic fluid, rotor brake oil, and windshield washer fluid.

Gross mass is the sum of the basic empty mass and the pilot/crew, the passengers, the baggage/cargo and the fuel. This value will vary with each mission.

6.1.2 Balance Definitions

Reference datum (RD) is an imaginary vertical plane located 3000 mm (118.1 in) forward of the Leveling Point (LP). For balance purposes, all horizontal distances (ARM) are measured from the RD (see Fig. 6-1).

Arm, for balance purposes, is the horizontal distance from the RD to the center of gravity of a given item. For other purposes, Fuselage Stations may be used. Fuselage stations are vertical planes perpendicular to, and measured along, the longitudinal axis of the helicopter. For the BO 105 helicopter arm and fuselage station are the same.

Moment (Massmoment) is the mass of an item multiplied by its arm.

Moment (kgmm) = Mass (kg) x Arm (mm)

Center of Gravity (CG) is the point about which the helicopter would balance if suspended. Distance from the RD is found by dividing the total moment by the gross mass of the helicopter.

$$\text{Arm (mm)} = \frac{\text{Sum of all moments (kgmm)}}{\text{Sum of all masses (kg)}}$$